

1. A causal LTI system whose system function is $H(z) = \frac{6 + 12z^{-1} + z^{-2} + 2z^{-3}}{4 - 3z^{-1} + z^{-2} - 2z^{-3}}$. Draw the signal flow graphs for implementations of the system in each of the following forms.
 - (a) Direct form II. (8%)
 - (b) Cascade form using first- and second-order direct form II sections. (8%)
 - (c) Parallel form using first- and second-order direct form II sections. (8%)
2. A Causal LTI system has a system function $H(z) = \frac{0.38}{1 - 1.87z^{-1} + 0.995z^{-2}}$. Let the coefficients be quantized with 7 bits ([7.5]). Please write the quantized system function $\hat{H}(z)$. Is this system $\hat{H}(z)$ stable? (10%)
3. Let $x_1[n] = \delta[n] - \delta[n-1] + 3\delta[n-2] - 2\delta[n-3] + 2\delta[n-4] - \delta[n-5]$ and $x_2[n] = \delta[n] + \delta[n-1] - 2\delta[n-3]$. Find their N -point circular convolution for $N=6$ and for $N=8$. (10%)
4. Draw the flow graph of the decimation-in-frequency 8-point FFT with output in normal order and input in bit-reversed order. (8%)
5. An ideal discrete-time highpass filter with passband cutoff frequency $\omega_c = 2\pi/5$ was designed using the (a) bilinear transform (b) impulse invariance with $T=2$ ms. What was the passband cutoff frequency Ω_c for the prototype continuous-time ideal highpass filter? (8%)
6. A causal LTI discrete-time system has system function $H(z) = \frac{(3 + 8z^{-1} - 3z^{-2})}{(2 - 9z^{-1} + 4z^{-2})}$. Find expressions for a minimum-phase system $H_1(z)$ and an all-pass system $H_{ap}(z)$ such that $H(z) = H_1(z)H_{ap}(z)$. (10%)
7. Let $X[k]$ denote the N -point DFT of the N -point sequence $x[n]$. Show that

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2. \quad (10\%)$$
8. Consider the product $(A+jB)(C+jD) = (AC-BD) + j(BC+AD)$, where A , B , C , and D are real

number. In this form, a complex multiplication requires four real multiplications and two real additions. Verify that a complex multiplication can be performed with three real multiplications and five real additions. (10%)

9. Suppose $x_1[n]$ is an infinite-length, stable sequence with z -transform given by $X_1(z) = 1/(1-(1/4)z^{-1})$. Suppose $x_2[n]$ is a finite-length sequence of length N , and the N -point DFT of $x_2[n]$ is $X_2[k] = X_1(z) \Big|_{z=e^{j2\pi k/N}}$, $k=0, 1, \dots, N-1$. Determine $x_2[n]$. (10%)